DA4 Assignment: 2: Panel data exercise [individual, 12p]

- V2.3

CO_2 emission and GDP

- To what extent does economic activity cause CO_2 emission?
- Importance
  - CO_2 emission may be the most important channel through which human activity leads to climate change

- Two flow variables
  - GDP: produced goods and services in a country over a year
  - CO_2: produced CO_2 in a country over a year

CO_2 emission and GDP Data

- World Development Indicators
  - Time series by country, 1992 to last available year
- GDP per capita
  - Measured in PPP USD at constant prices
- CO2 emission per capita
  - Get total emission and divide by population
- Think about functional form

Estimate and interpret

- Run these models and interpret coefficients for each of them

1. Cross-section OLS for the year 2005
2. Cross-section OLS for the last year
3. First difference model, with time trend, no lags
4. First difference model, with time trend, 2 year lags
5. First difference model, with time trend, 6 year lags
6. Fixed effects model with time and country fixed effects

Tasks

- Download data and describe it (1p)
- Consider coverage (missing values), drop countries with poor coverage document it (1p)
- Estimate and discuss coefficients for each model (4p)
- Think of a potential confounder, add to models 1,4,6 and discuss your finding (2p)
- Summarize your finding in a paragraph (2p)
- Make sure it’s a high quality, pretty submission (2p)

To submit

1. Submit a pdf report
   1. could be markdown .html
   2. NOT a notebook
      1. *It must not contain useless stuff*
2. Code and data should be either
   1. submitted
   2. Link provided / hosted in public repo such as github

AI policy

- You can use AI to
  - Write code + help understand code
  - Formatting, making graphs, tables pretty

- I'll expect a high quality very pretty submission